



EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Spring 2026



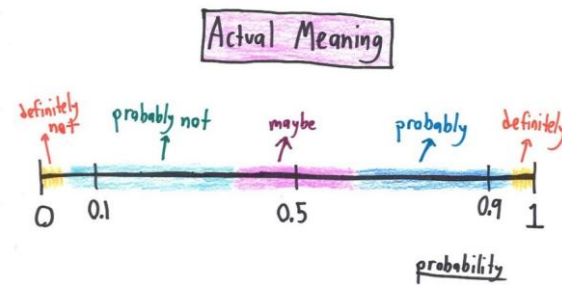


EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

Aamir Hasan

Week 13 - Lecture No 25 – April 08, 2026



Announcements!

1. Office hours (post Ramadan) – Tuesday & Thursday 12:00 to 1:00 PM
2. Planned quizzes
 1. Quiz No 08 – April 15, 2026 (Wednesday)
 2. Quiz No 09 – April 22, 2026 (Wednesday)
 3. Quiz No 10 – April 29, 2026 (Wednesday)

Agenda for today

1. Quiz No 07
2. Unit 5 – Continuous Random Variable
 - Jointly Continuous distributions

Quiz No 07

20:00

Jointly continuous R.V.'s and Joint PDFs.

$$p_X(x) \quad f_X(x)$$

$$p_{X,Y}(x, y) \quad f_{X,Y}(x, y)$$

$$p_{X,Y}(x, y) = \mathbf{P} (X = x \text{ and } Y = y) \geq 0$$

$$f_{X,Y}(x, y) \geq 0$$

$$\mathbf{P}((X, Y) \in B) = \sum_{(X, Y) \in B} \sum p_{X,Y}(x, y)$$

$$\mathbf{P}((X, Y) \in B) = \iint_{(X, Y) \in B} f_{X,Y}(x, y) \, dx dy$$

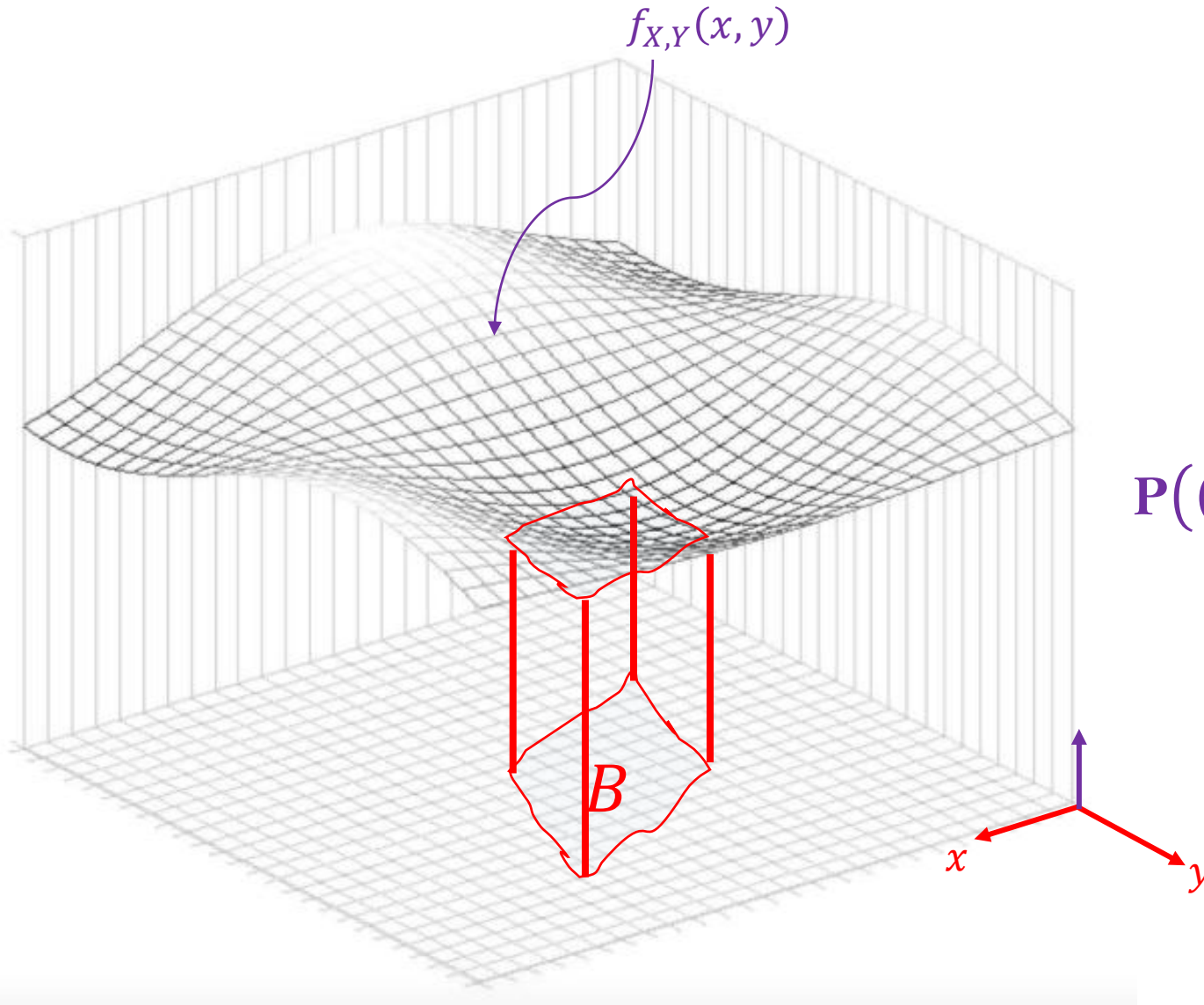
$$\sum_x \sum_y p_{X,Y}(x, y) = 1$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) \, dx dy = 1$$

Definition:

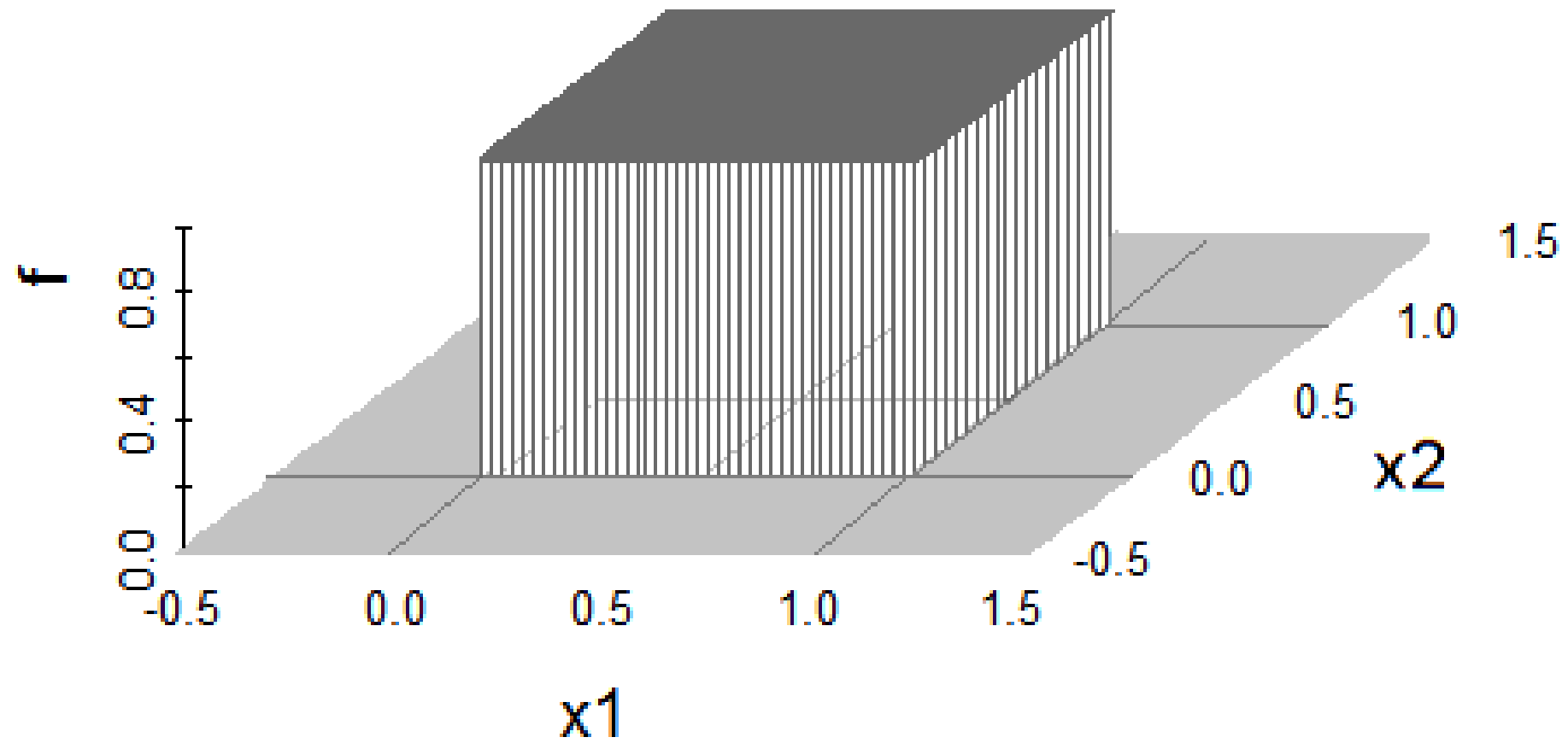
Two random variables are **jointly continuous** if they can be described by a joint PDF

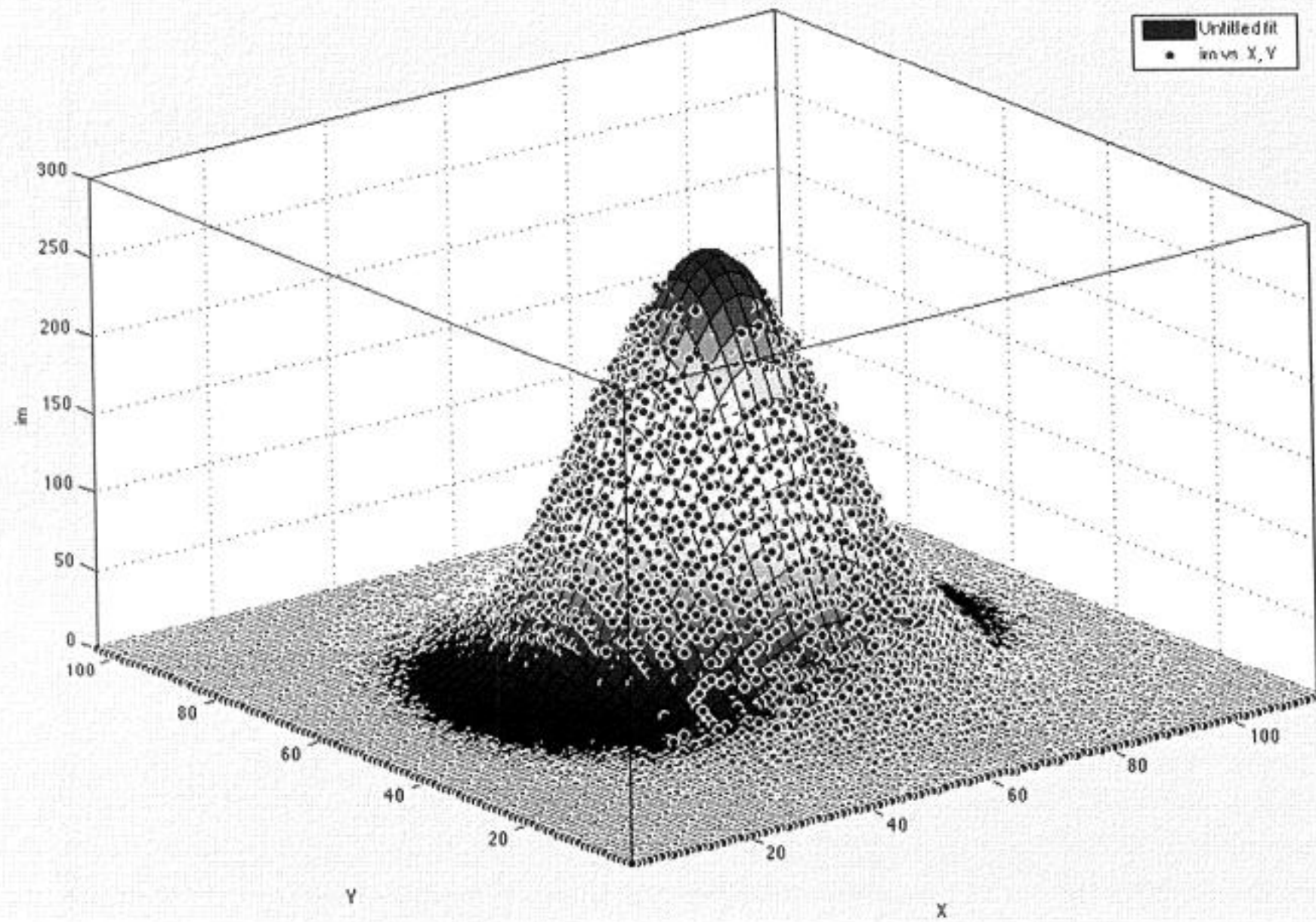
Visualizing a Joint PDF



$$\mathbf{P}((X, Y) \in B) = \iint_{(X, Y) \in B} f_{X,Y}(x, y) \, dx dy$$

bivariate uniform





On Joint PDFs

$$\mathbf{P}((X,Y) \in B) = \iint_{(X,Y) \in B} f_{X,Y}(x,y) dx dy$$

$$\mathbf{P}(a \leq X \leq b, c \leq Y \leq d) = \int_c^d \int_a^b f_{X,Y}(x,y) dx dy$$

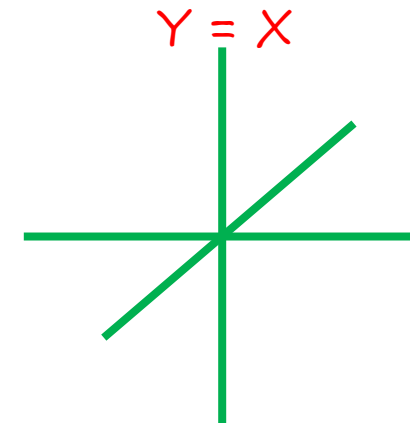
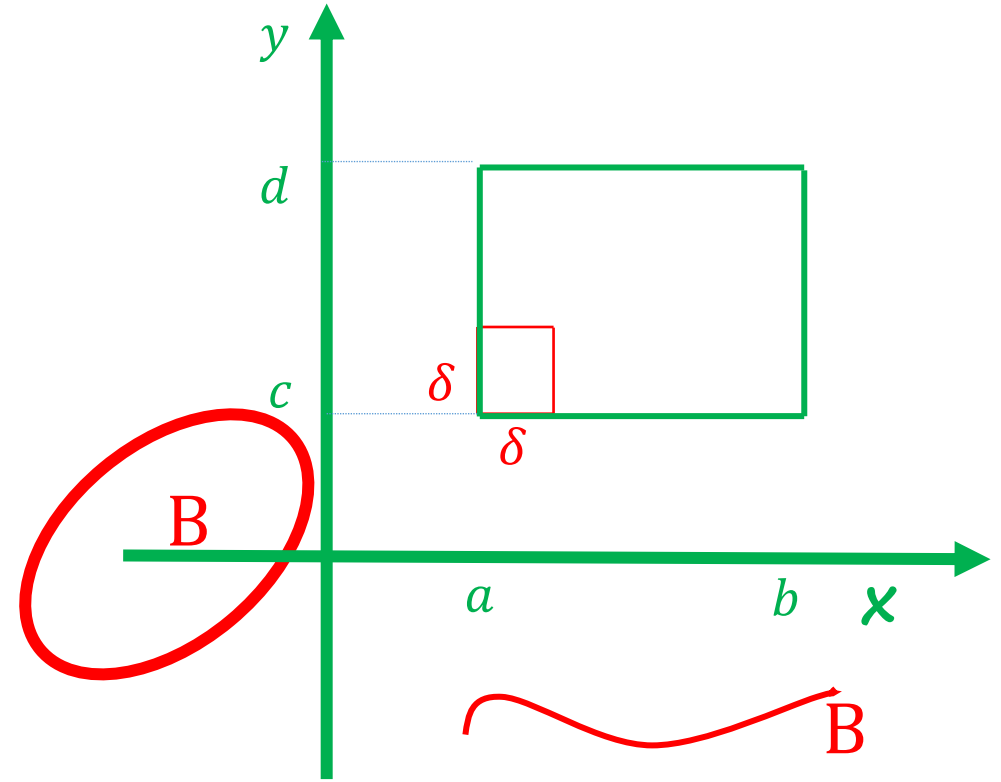
$$\mathbf{P}(a \leq X \leq a + \delta, c \leq Y \leq c + \delta) \approx f_{X,Y}(a, c) \delta^2$$

$f_{X,Y}(x,y)$: probability per unit area
around the point (x,y)

$$\text{area}(B) = 0 \Rightarrow \mathbf{P}((X,Y) \in B) = 0$$

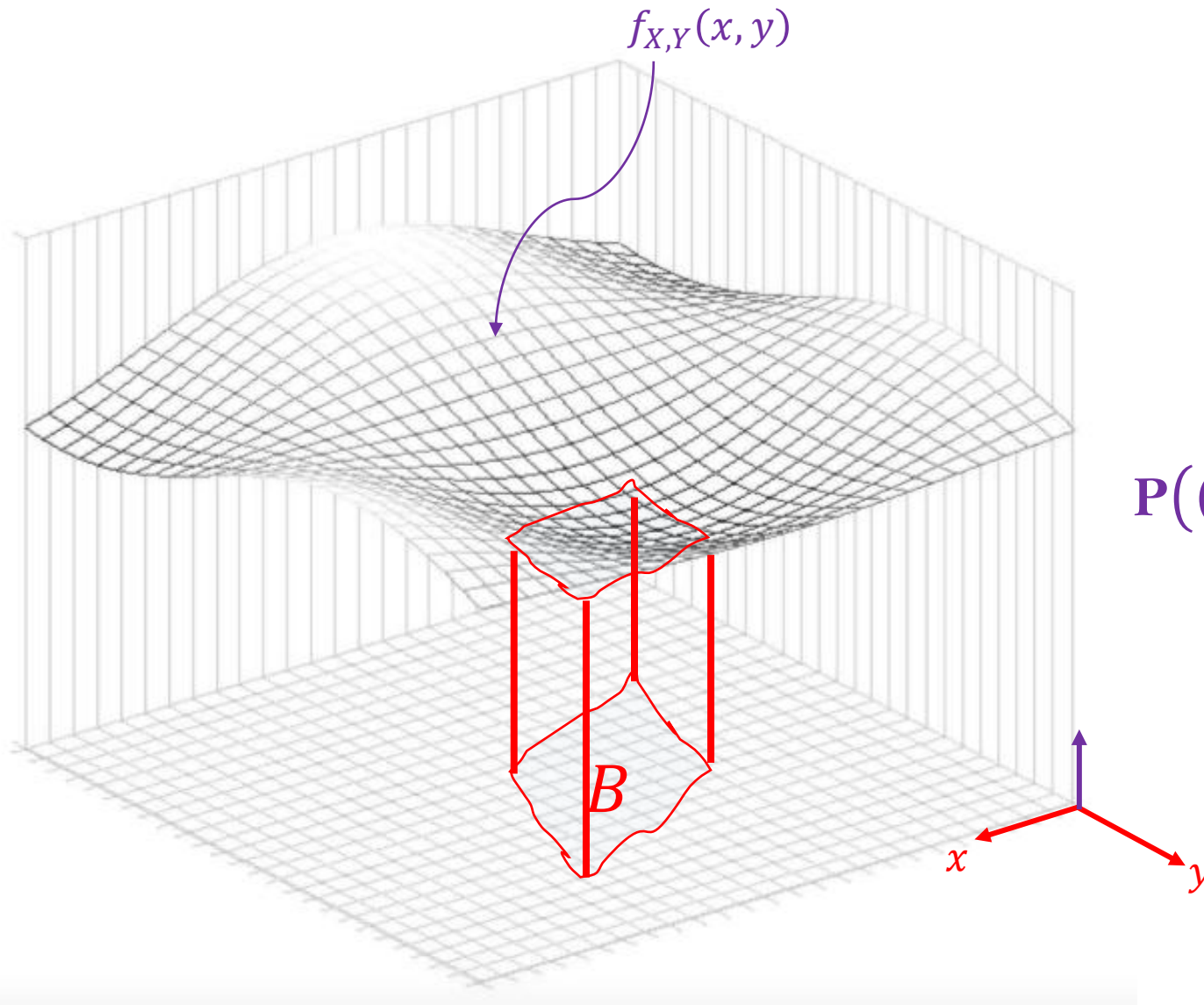
Joint PDF

area



Are X and Y jointly
continuous?

Summary - Joint PDF



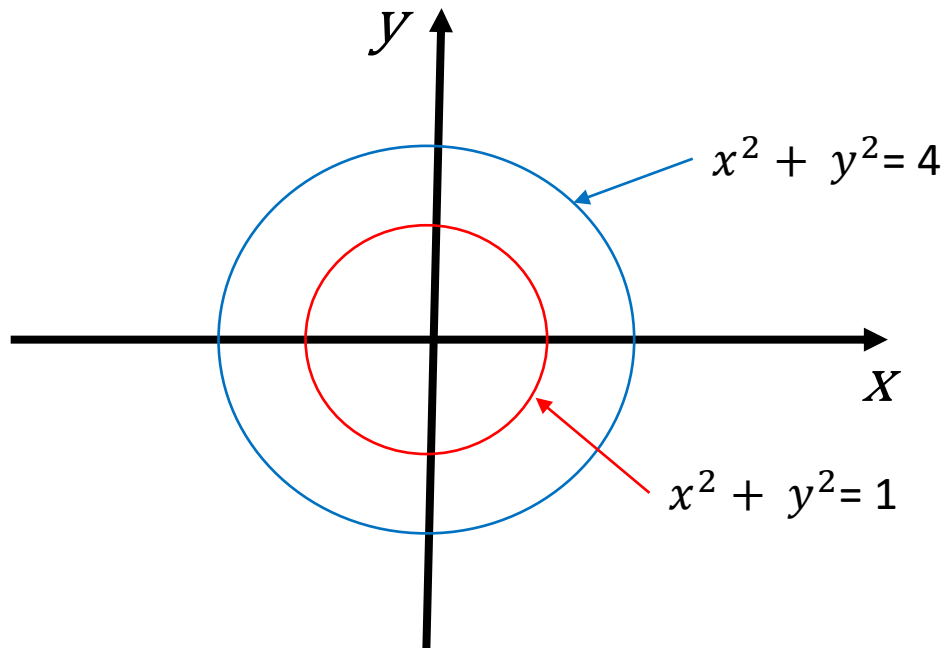
$$\mathbf{P}((X,Y) \in B) = \iint_{(X,Y) \in B} f_{X,Y}(x,y) \, dx dy$$

Independent Standard Random Variables

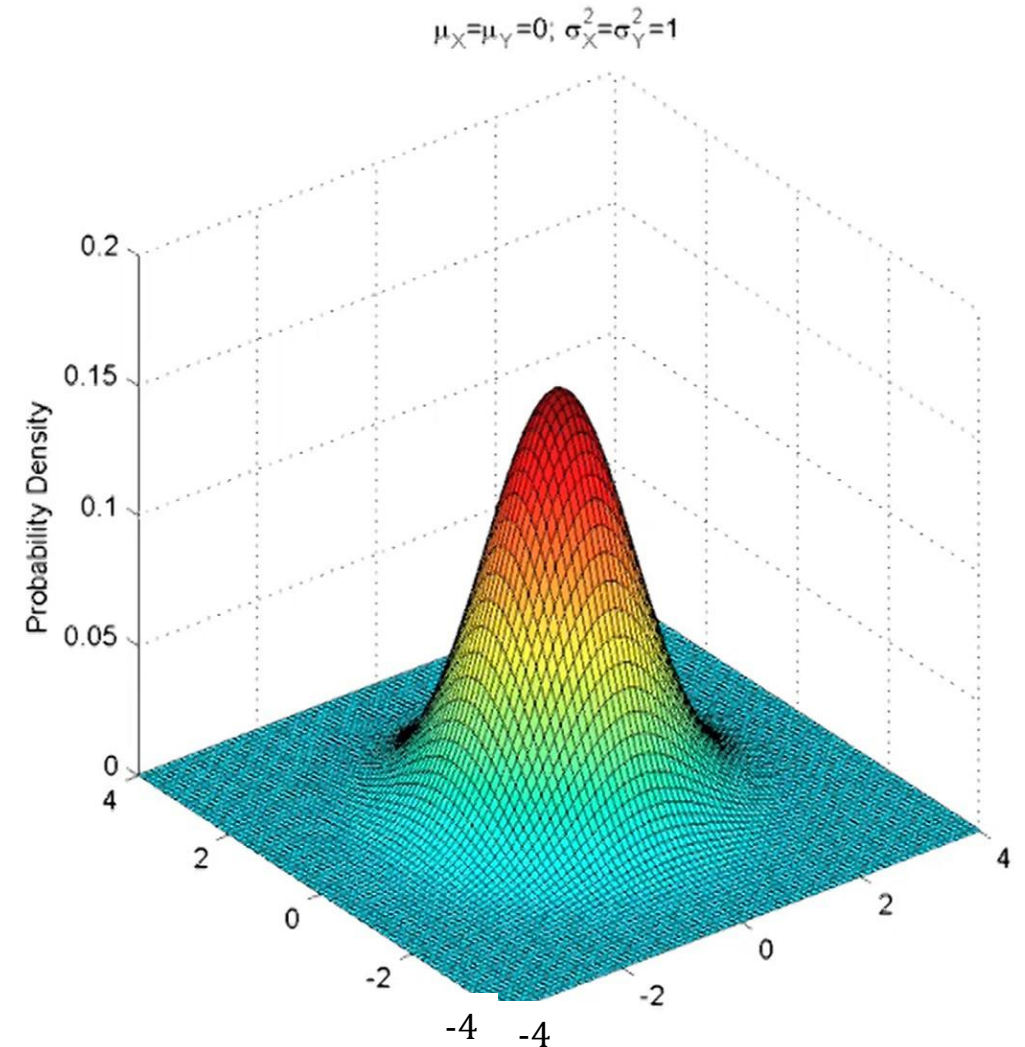
$$f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y)$$

$$= \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\} \cdot \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{y^2}{2}\right\}$$

$$= \frac{1}{2\pi} \exp\left\{-\frac{1}{2}(x^2 + y^2)\right\}$$

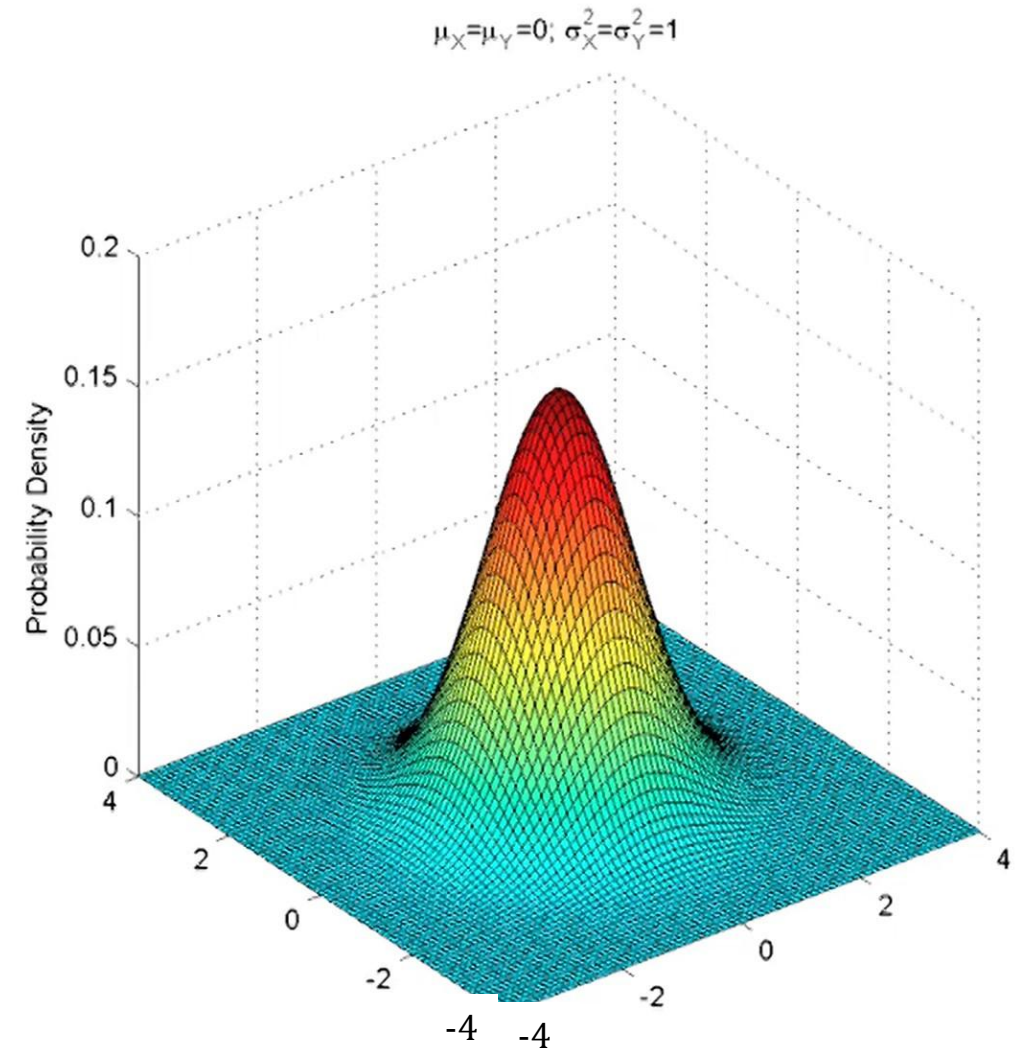
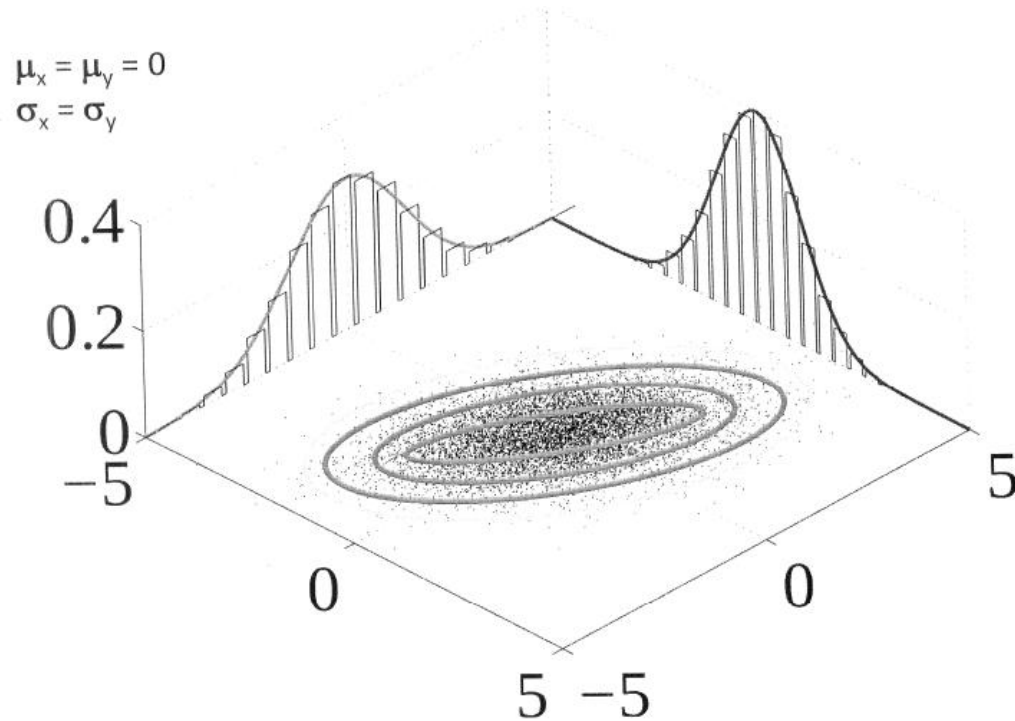


- General normal $N(\mu, \sigma^2)$: $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$



Independent Standard Random Variables

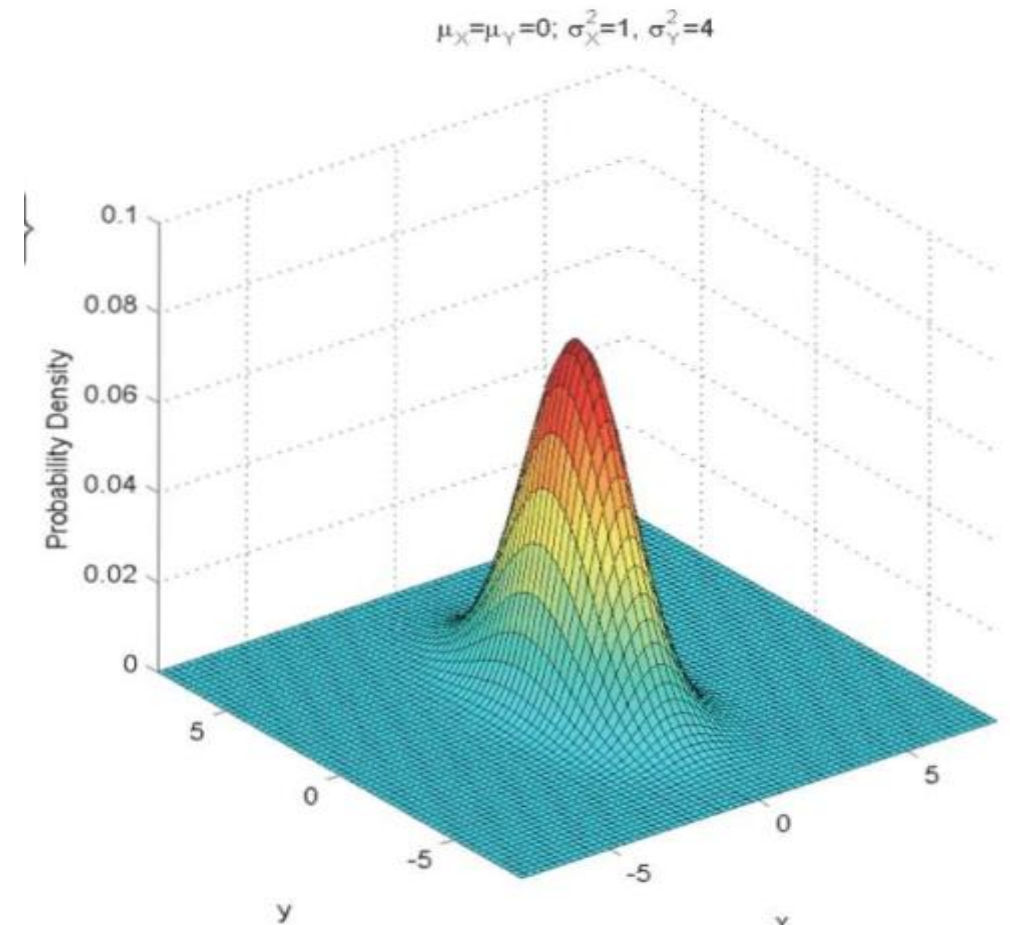
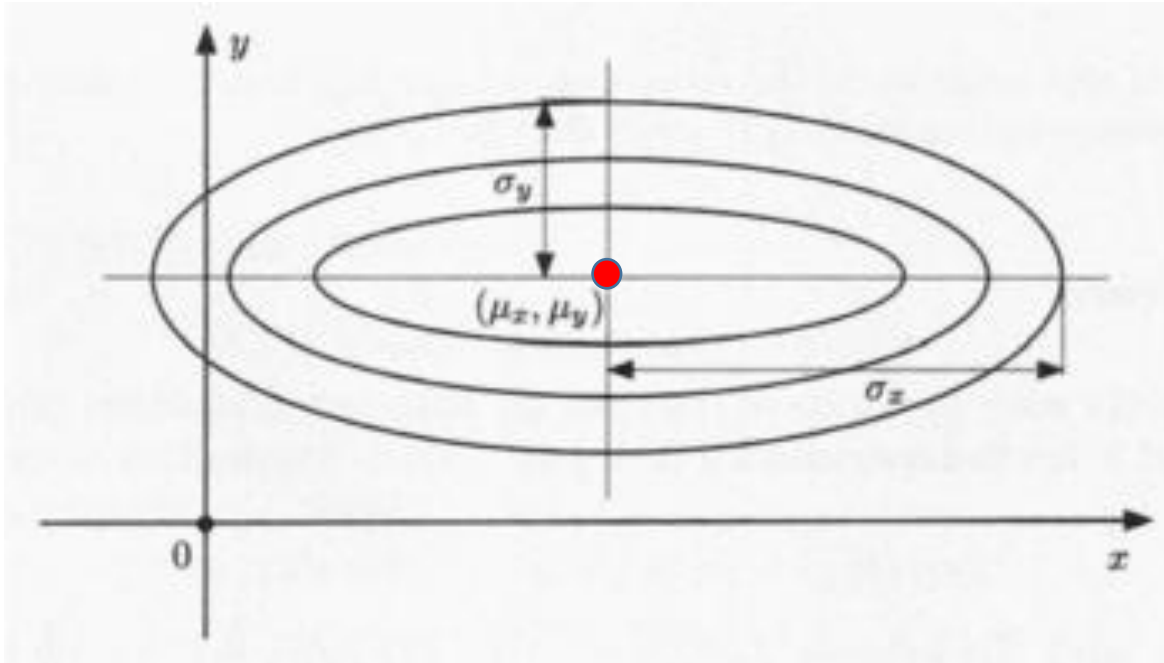
$$\begin{aligned} f_{X,Y}(x,y) &= f_X(x) \cdot f_Y(y) \\ &= \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\} \cdot \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{y^2}{2}\right\} \end{aligned}$$



Independent Normal Random Variables

- General normal $N(\mu, \sigma^2)$: $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$
$$= \frac{1}{\underline{2\pi\sigma_x\sigma_y}} \exp\left\{-\frac{(x-\mu_x)^2}{2\sigma_x^2} - \frac{(y-\mu_y)^2}{2\sigma_y^2}\right\}$$



Bivariate Normal Random Variables

$$f_X(x) = \frac{1}{\sigma_x \sqrt{2\pi}} e^{-\frac{(x-\mu_x)^2}{2\sigma_x^2}}$$

$$f_Y(y) = \frac{1}{\sigma_y \sqrt{2\pi}} e^{-\frac{(y-\mu_y)^2}{2\sigma_y^2}}$$

$$f_{X,Y}(x,y)$$

